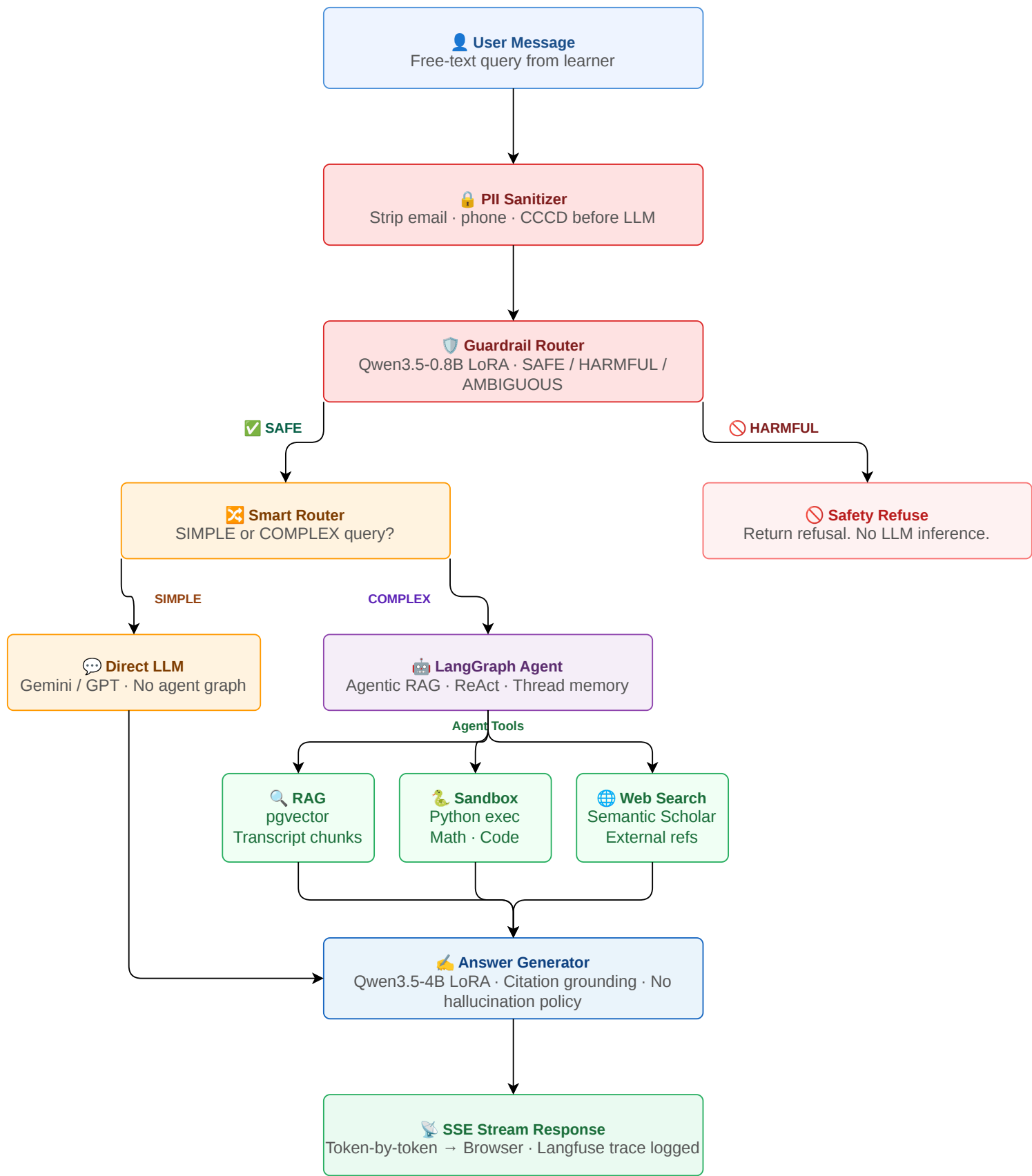
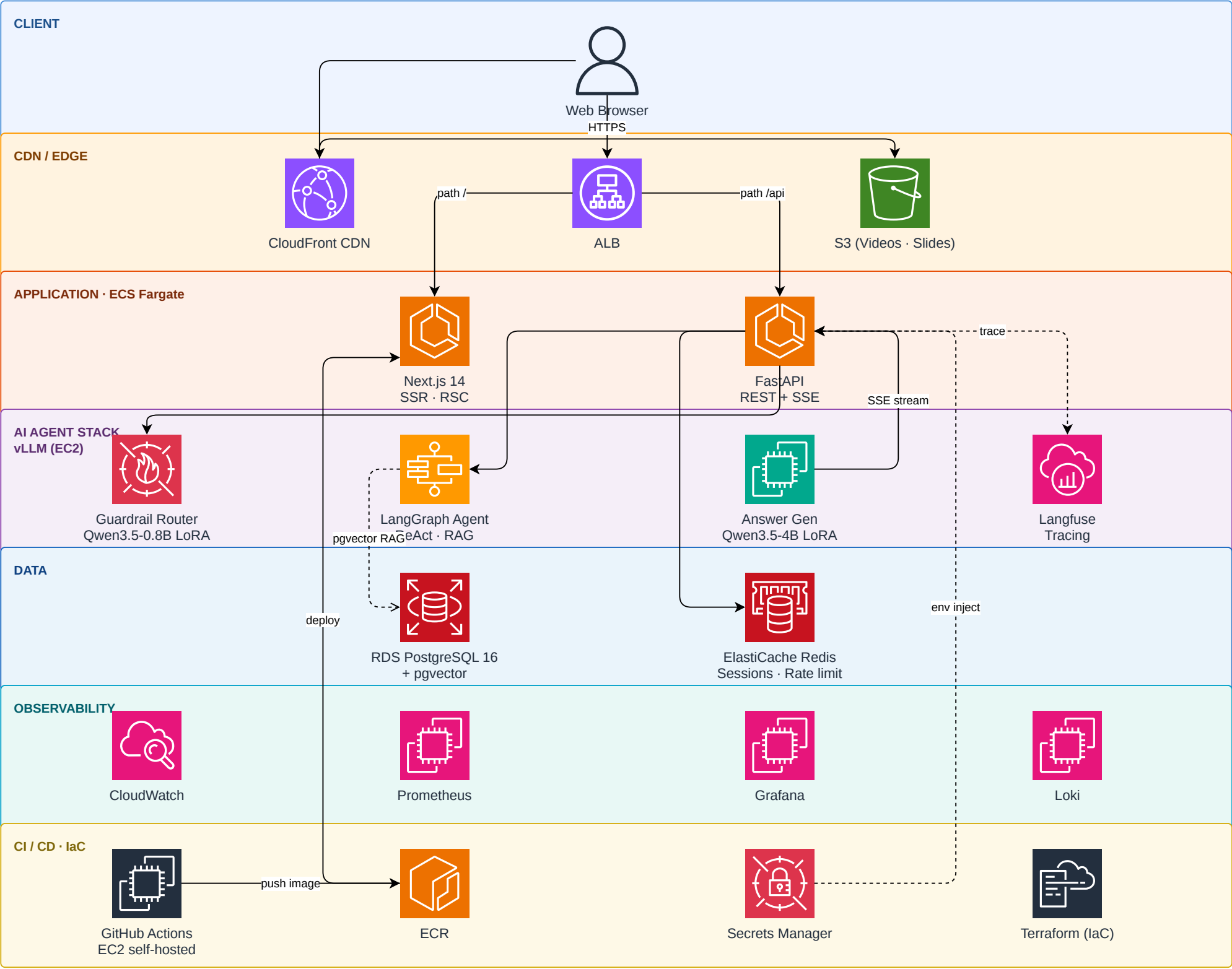
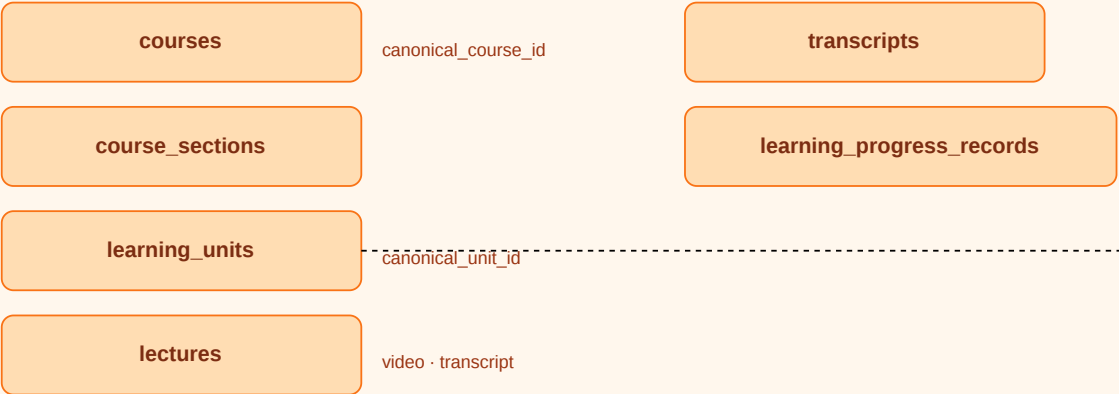


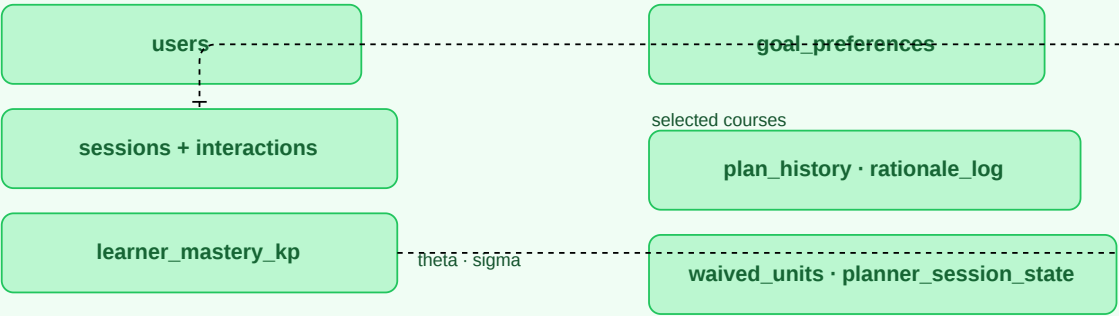




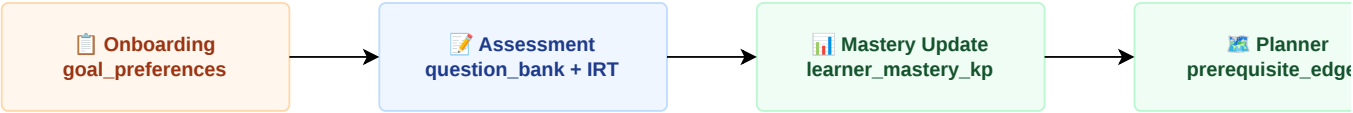
 Layer 1 — Product Shell
Course catalog & content structure



 Layer 3 — Learner State
Mastery · Planner · Session tracking

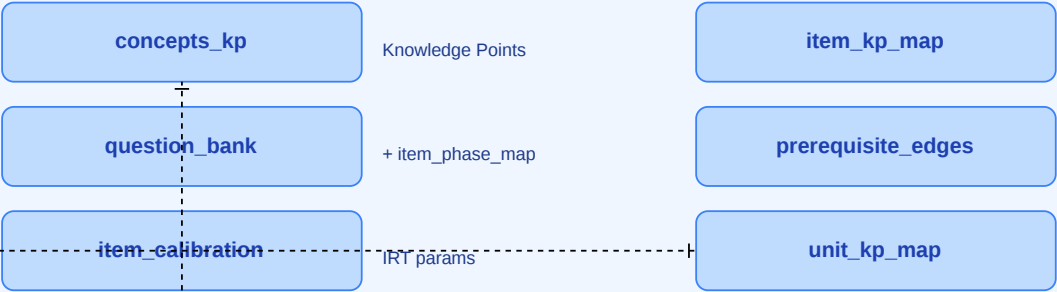


Adaptive Learning Loop — I

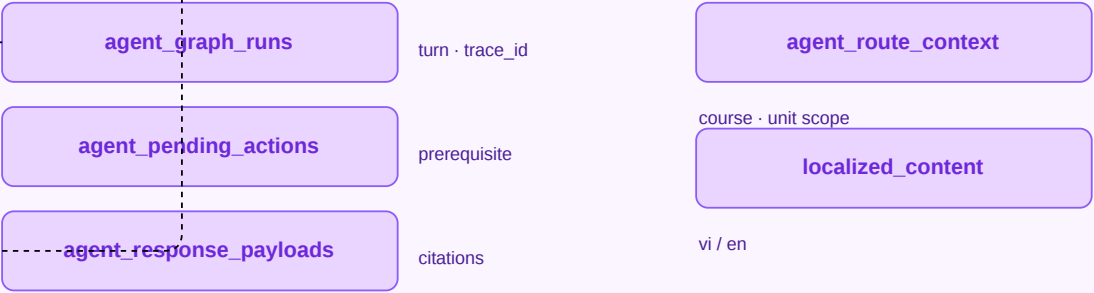


PostgreSQL 16 + pgvector

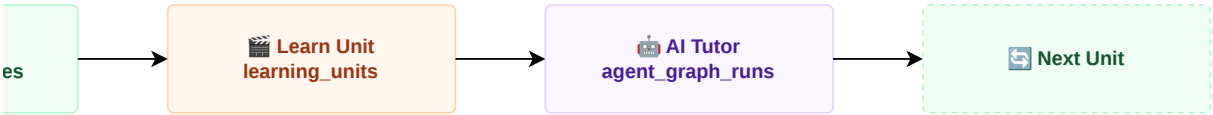
Layer 2 — Canonical Content
graph · Assessment · Prerequisites



Layer 4 — Agent State
gGraph run state · Citations · Actions



How layers interact



User Onboarding & Assessment Flow

Phase 1 — Goal & Topic Discovery

Select Goals
Choose learning objectives

Browse Topics
Explore courses & units

Mark Known Topics
Skip what you already know

Phase 2 — Placement Assessment

Start Assessment
System generates questions
per topic (Easy · Medium · Hard)

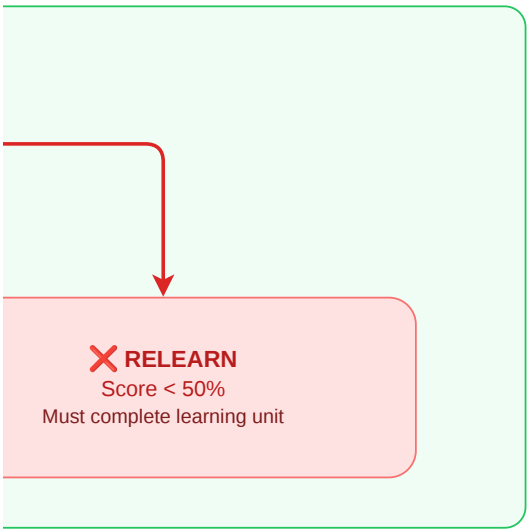
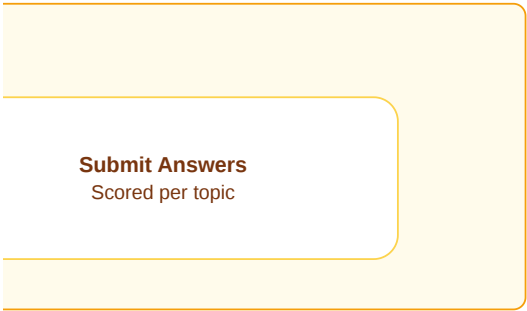
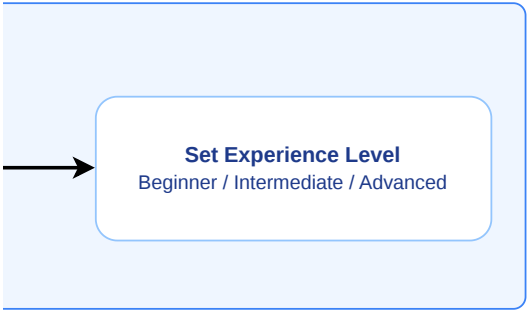
Answer Questions
Multiple choice · local state

Phase 3 — Results & Decisions

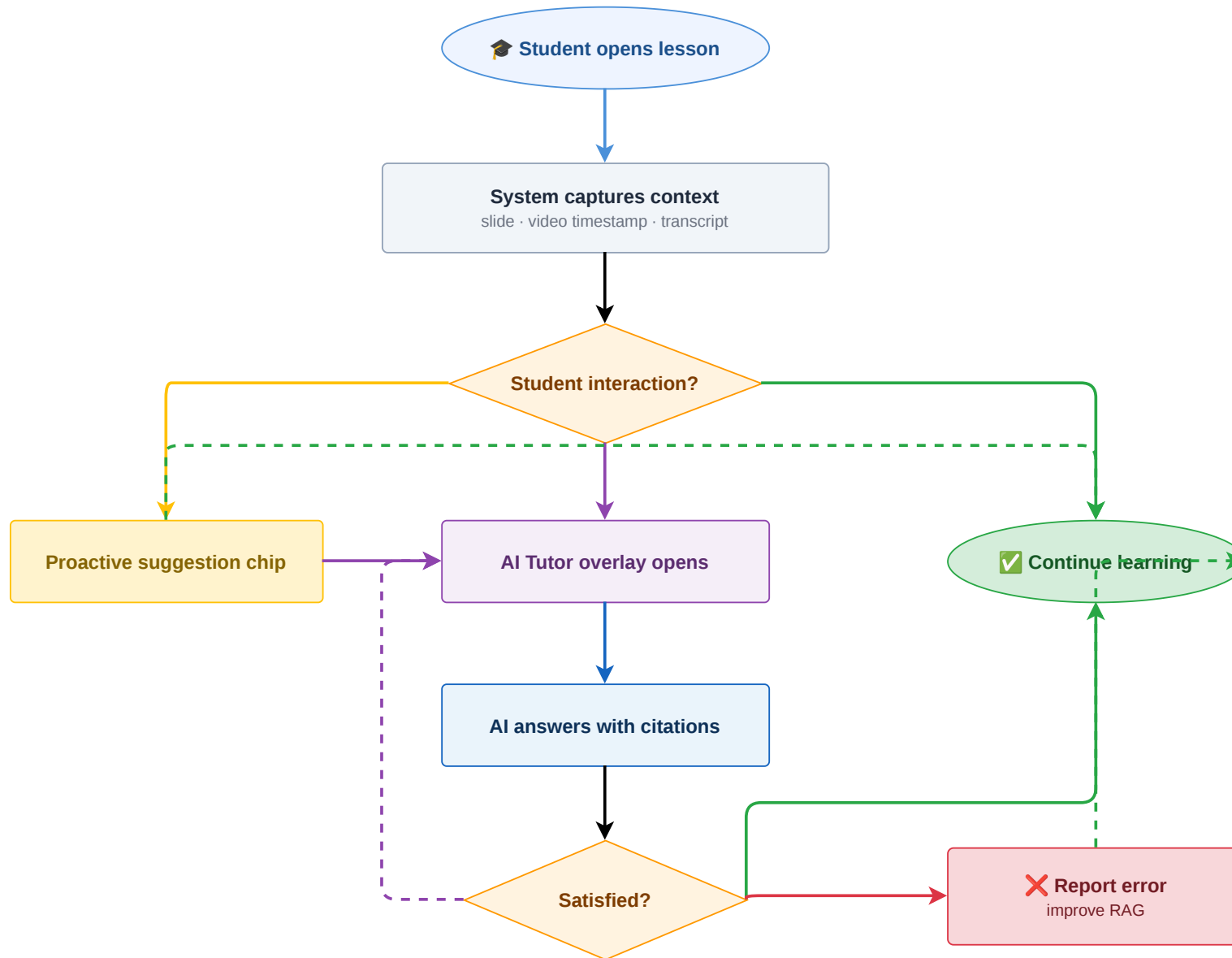
View Results
Score per topic

✓ **SKIP**
Score $\geq 70\%$
Move to next topic

⚠ **REVIEW**
50% – 69%
User chooses: skip or relearn



UX Flow — AI Tutor Overlay



Data Flywheel — Continuous Improvement Loop

